



Simon Pelletier

CEP | M.A.Sc. Candidate
R&D | Mechanical Engineering

✉ Lévis & Montréal, Québec

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in SimPelletierPouliot

🌐 SimPelletier

🌐 Link to Online Resume

About me

I am a mechanical engineering candidate with a specialization in mechanical design. I possess particular expertise in the use of Python and open-source libraries for the design and validation of mechanical structures, which I have applied to my master's project.

My experience includes mechanical design of composite structures, manufacturing processes, tooling and machine element design. Additionally, I have a strong understanding of cable mechanics, which I acquired as part of my master's research.

Skills

Mechanical Design (SolidWorks, Catia, Fusion360)

Programming (Python, Numba, Latex)

Structural Engineering (Ansys APDL, Altair, Fatigue)

Data Analysis (Numpy, Pandas, Sk-learn)

Computer Vision (OpenCV)

Scientific Visualization (VTK, Paraview)

French(Native) English(Professional)
Spanish(Beginner) Solution-Oriented

Experience

- Since 2025 **AEOLUS Project Manager**
Preformed Line Products (PLP)
Power Line Motion Monitoring Systems
- Design of new monitoring devices
 - Installations coordination
- 2022 to 2024 **Practical class teacher for MEC528 course** Part-time
École de Technologie Supérieure (ETS)
- Synthesize concepts related to machine elements and fatigue.
 - Carry out classroom and laboratory exercises.
- 2020 to 2022 **CEP Project Manager**
Centre de développement des composites du Québec (CDCQ)
- Design of tooling and structural composite parts for high-performance and lightweight solutions.
- Sum 2019 **Mechanical Designer** Internship
Rocky Mountain Bicycles
- Design of reliability test benches for electric motor components.
 - Design and implementation of an electric motor efficiency bench test.
 - Post-processing of test data.
 - Development of test procedures.
- Sum-Fal 2017 **Mechanical Designer** Internship
Centre de développement des composites du Québec (CDCQ)
- Design of tooling and structural composite parts.
 - Implementation of a large-capacity FDM 3D printer.
 - Design of a nylon in-situ curing machine.
- Sum 2016 **Draftsman**
Les Équipements Marki

Education

- 2022 to 2025 **M.A.Sc. candidate in Mechanical Engineering**
To École de Technologie Supérieure (ETS)
Comprehensive mechanical modeling of cables under cyclic loading for service life prediction.
- 3D reconstruction of cable internal wire paths from computed tomography data using image processing (*OpenCV*). 🌐
 - Macro-scale modeling of internal cable structures using homogenization approaches (*Python*, *APDL*).
 - Wire-level modeling of a braided structure inside cable outer layers with several million contact points.
 - Analysis of contact interaction data using open-source 3D scientific visualization tools (*VTK*).
- Win-Spr 2020 **Abroad Studies**
At Universitat Politècnica de Catalunya (UPC)
- 2016 to 2020 **B.Sc. in Mechanical Engineering**
From École de Technologie Supérieure (ETS)
- 2013 to 2016 **College in Mechanical Engineering**
From Cégep Lévis-Lauzon

Projects

- Spr 2022 **Custom Linear FEA Program of Composite Beams and Plates**
École de Technologie Supérieure (ETS-SYS806)
- Orthotropic properties derived from the Classical Laminate Theory.
 - Beam element formulation based on Timoshenko beam theory.
 - Plate element formulation based on Reissner-Mindlin theory.
 - Validated with Ansys® software using the ACP module.
- Win-Spr 2020 **Design of a Bicycle Cable Disc Brake** 🌐
Universitat Politècnica de Catalunya (UPC)
- 2016 to 2018 **Participation in the Student Project C-Class Catamaran** 🌐
École de Technologie Supérieure (ETS)
- Spr 2016 **Design and Manufacture of a Nylon Rope Machine Cutter** 🌐
From Cégep Lévis-Lauzon
- Design and manufacture of the prototype.
 - Program an Arduino board for system operation and user interface.

Awards

- 2023 **Recipient of an Excellence Scholarship (ETS).**
- 2018 & 2020 **Recipient of Two Scholarships from Private Companies to Encourage Career Choice (ETS).**